FLEET TEAM DIGEST

737MAX-FTD-28-17002

Issue Title : Right Main Tank Fuel Spill

Airplane Model	ATA	Minor Model(s)
737MAX	2821-00	-7, -8, -8200, -9, -BBJ
Other Model(s)	**ECCN**	
	9E991	

Originated Date	Last Revised Date	Created On
07/13/2017	07/13/2017	01/26/2017
Estimated Completion Date	**Next Update Date**	

Description

The purpose of this FTD article is to provide suggested refueling actions to minimize a potential fuel spill from the right-hand (RH) tank during fueling operations.

These refueling actions were based on reported best practices from operators and have not been tested or verified by Boeing.

Background

Fueling is complete when the deadman switch is released and the refuel switches are set to CLOSE. Additionally, the shutoff valves are closed automatically when the refuel float switch reaches full in each tank to prevent an overfill. The manual override button will open the refuel valves and disable the automatic shutoff for the tank being refueled, which can contribute to an overfill and subsequent fuel spill.

At the fueling station, the Fuel Quantity Indication System (FQIS) does not control the fueling valve but simply provides fuel quantity indication. Volumetric Top Off (VTO) is provided by the refuel float switch. If the FQIS senses the tank is over full as calculated by the computer, it will cause the refuel gauge to flash.

Fuel temperature impacts the calculation of fuel quantity by the FQIS. Extra time is needed for fuel mixing and can result in errors in the indicated fuel in the RH tank during fueling.

As old and new fuel are mixed with different temperatures, it may take time for the temperature to stabilize. During this time, the fuel quantity calculation by the FQIS may not be accurate. If the gauge flashes, the fueler should stop fueling per the AMM procedure. A flashing gauge indicates that a fuel spill is imminent. Because of the location of the refuel discharge in the RH main tank, 7 bays outboard from the compensator, with extremely cold fuel in the tank and warm fuel being added on top, it is possible that the gauge will not flash before a fuel reaches the refuel float switch, which is the last defense against a fuel spill.

The RH tank fuels slightly faster than the left-hand (LH) tank. RH tank refuel tubing is much shorter and the orifices do not perfectly balance the fuel flow rates. The LH tank does not have the stratification issue because the fuel is discharged into the same bay as the compensator near the bottom of the tank so the fuel is mixed faster. The center tank does not have the issue because it is usually empty when refueling so there is no cold fuel in the center tank.

Status

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No updates planned to the fuel system. Best practices for refueling the aircraft, outlined in this FTD article, adequately address the issue of fuel stratification leading to right main tank overfill during refueling procedures.

Interim Action

See the operator action section below for manual override button use recommendation to avoid fuel spills.

Final Action

Fuel the RH tank part way, approximately 80%, and then wait approximately 6 to 7 minutes to let the cold and warm fuel mix enough for the indication to be more representative of the actual fuel quantity in the tank. During this time, the LH tank may be fueled while waiting for the fuel to mix in the RH tank.

Follow recommended operator action, as outlined below, while fueling the airplane.

Operator Action

1. Do not use the Manual Override button unless there is a known FQIS issue, and fuel quantity is detected using other means (fuel measuring sticks).

a. There is a possibility of the button sticking(See FTD 737NG-FTD-28-12004).

2. If a refuel gauge starts to flash, release the "deadman" switch immediately to prevent a fuel spill.

a. The flashing gauge indicates a tank which is volumetrically full as calculated by the computer.

b. The refuel valve can fail to close or the float switch can fail to open resulting in over filling the tank and eventually a fuel spill.

Note: If there is fuel temperature stratification (cold fuel at the compensator and warm fuel added on top) it is possible that the gauge will not flash before a fuel spill occurs.

3. When the system automatically shuts off (float switch activation), do not top-off the tank.

a. Possibility that the RH side will experience fuel temp stratification and read up to 20 kg (44 lbs) low.

b. The RH side fills slightly faster than the LH side.

c. It is acceptable to have each tank automatically shut off at slightly different quantities.

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Fleet Team Digest Article (FTD)	737NG-FTD-28-12004		

Related Categories

- Airplane Maintenance Issues : Issues that occur during airplane maintenance or have an impact on airplane reliability
- Best Practices / Operational Efficiency : Operational / Maintenance-focused best practices, techniques, health checks or recommendations for preserving aircraft efficiency.
- Fleet Team IdeaXchange : Topic of discussion on the Fleet Team IdeaXchange bulletin board

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- In Service Occurrences : null

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